

UGANDA CHRISTIAN UNIVERSITY

A Centre of Excellence in the Heart of Africa

COURSE: **MASTER OF SCIENCE IN DATA SCIENCE AND ANALYTICS**

COURSE UNIT: **DATA MINING MODELING AND ANALYTICS**

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ASSIGNMENT ONE PROJECT TWO REPORT

Data Exploration

The data has 9 variables, 8 input variables and 1 output variable. The output variable is concrete compressive strength measured in MPa. The input variables are cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate and age. The target is compressive strength.

Dataset characteristic	Finding
Original observations	1,030
Columns	9 (8 inputs and 1 output)
Missing values	0 in every column
Exact duplicate rows	25
Observations after duplicate removal	1,005
Mean compressive strength (original data)	Approximately 35.82 MPa
Minimum compressive strength	Approximately 2.33 MPa
Maximum compressive strength	Approximately 82.60 MPa

All variables are numeric. This makes the dataset suitable for regression modelling without categorical encoding. The strength values cover a broad range, indicating that a model must perform across both low- and high-strength concrete observations.

Data Preparation and Justification

Preparation step	Treatment and justification
Column names	Trimmed leading/trailing whitespace from column names so the documented target and feature names can be referenced reliably.
Missing values	No missing values were observed. Median imputation was nevertheless included in the model pipelines as a defensive preprocessing step; with this dataset it does not change complete observations.
Duplicate records	Removed 25 exact duplicate rows, leaving 1,005 records. Exact duplicates can give identical observations repeated influence. Other repeated or similar mixtures were not removed unless all fields matched.
Feature and target separation	Used the eight documented inputs as predictors and concrete compressive strength (MPa) as the target, following the supplied README.
Train/test split	Used an 80:20 random split with random_state=42 (804 training and 201 test observations) to assess generalisation on held-out observations.
Scaling	Standardised numerical inputs for Linear Regression's pipeline. Tree-based models do not require standardisation,

	so their pipeline uses median imputation without scaling.

Mean Prediction Baseline

A DummyRegressor using strategy='mean' was fitted on the training data. It predicts the training-set mean compressive strength for every test observation. This provides a simple benchmark: useful models should reduce MAE and RMSE and achieve a higher R^2.

Baseline metric	Test result
MAE	13.9723 MPa
RMSE	17.2965 MPa
R^2	-0.0028

The negative R^2 indicates that, on this test split, constant mean prediction performed slightly worse than predicting the mean of the observed test outcomes. The baseline has no ability to respond to changes in mixture composition or curing age.

Linear and Nonlinear Model Comparison

Linear Regression was used as the linear approach. Random Forest Regressor and HistGradientBoostingRegressor were selected as nonlinear tree-ensemble approaches because they can learn nonlinear relationships and interactions among ingredients and age. All models were trained on the same training partition and evaluated on the same held-out test partition.

Model	MAE (MPa)	RMSE (MPa)	R^2	Model
HistGradientBoosting	3.1144	4.6823	0.9265	HistGradientBoosting
Random Forest	3.7163	5.3642	0.9035	Random Forest
Linear Regression	8.8960	11.1922	0.5801	Linear Regression
Mean baseline	13.9723	17.2965	-0.0028	Mean baseline

HistGradientBoosting performed best, with the lowest MAE and RMSE and the highest R^2. Random Forest also performed strongly. Both nonlinear models substantially outperformed Linear Regression. The comparison supports the view that the relationship between the concrete ingredients, curing age and compressive strength is not adequately represented by a simple linear model in this dataset.

MAE expresses the average absolute prediction error in MPa. RMSE penalises larger errors more strongly. R^2 describes the proportion of test-set variation explained by the model relative to a mean predictor. The high R^2 of HistGradientBoosting indicates strong predictive performance on this particular split, but does not establish causal effects or guarantee performance on other concrete populations.

Residual Analysis and Weak Prediction Areas

Residuals were calculated as actual strength minus predicted strength for the best-performing model, HistGradientBoosting. A positive residual means the model underpredicted strength; a negative residual means it overpredicted strength. The test residual mean was approximately +0.6456 MPa, residual standard deviation was approximately 4.6376 MPa, and residual MAE was 3.1144 MPa.

Error by Curing Age

Age group	Test records	MAE (MPa)	Mean residual (MPa)	Age group
1–7 days	56	3.043	-0.185	1–7 days
8–28 days	89	3.505	+1.071	8–28 days
29–90 days	27	2.748	+0.772	29–90 days
91–365 days	29	2.393	+0.828	91–365 days

Among the age groups examined, predictions were weakest for observations aged 8–28 days, which had the highest age-group MAE of 3.505 MPa. Its positive mean residual (+1.071 MPa) indicates a tendency to underpredict strength in this group on average. The 91–365 day group had the lowest MAE (2.393 MPa), although this group contained only 29 test observations.

Error by Actual Strength

Actual-strength quartile	Test records	MAE (MPa)	Mean residual (MPa)	Actual-strength quartile
2.331–23.801 MPa	51	2.117	-0.818	2.331–23.801 MPa
23.801–34.570 MPa	50	2.663	-0.850	23.801–34.570 MPa
34.570–46.898 MPa	50	3.147	+0.702	34.570–46.898 MPa
46.898–82.599 MPa	50	4.551	+3.578	46.898–82.599 MPa

The weakest predictions occurred in the highest actual-strength quartile: its MAE was 4.551 MPa, the largest among the four groups. Its mean residual was +3.578 MPa, indicating systematic underprediction of high-strength observations on average. This is evidence that the model is less accurate at the upper end of the strength range. The actual-versus-predicted and residual plots in the accompanying code provide visual diagnostics for this pattern.

These subgroup results are descriptive and depend on the particular random test split. The number of records in some age groups is relatively small, so the differences should be interpreted cautiously.

Does Cement Association Prove a Real-World Effect?

Cement association doest prove a real-world effect. A predictive association between cement content and compressive strength does not prove what will happen if cement content is deliberately changed in a real concrete mixture. In this dataset, cement content has a Pearson correlation of approximately 0.4883 and a Spearman correlation of approximately 0.4614 with compressive strength. These values indicate a moderate positive association in the observed data, but they do not establish causation.

Concrete strength depends on several ingredients and their interactions, including water, blast furnace slag, fly ash, superplasticizer and aggregate quantities, as well as curing age. Cement content may also vary together with other mixture components. Consequently, a model may use cement as a predictive feature while its estimated association reflects the combined patterns in the observed mixtures. Changing cement alone may produce a different result from the model's observational prediction.

To estimate the causal effect of changing cement content, controlled experiments would be needed. For example, cement could be varied systematically while other mixture proportions and relevant conditions, such as curing conditions and testing procedures, are controlled. Replicated tests would help distinguish the effect of cement from random variation and the effects of other ingredients.

Conclusion

The concrete dataset contains 1,030 observations, with 25 exact duplicates removed and no missing values identified. The mean-prediction baseline performed poorly (RMSE 17.2965 MPa; R^2 -0.0028). Nonlinear models outperformed Linear Regression, with HistGradientBoosting achieving the best test performance (MAE 3.1144 MPa, RMSE 4.6823 MPa and R^2 0.9265). Residual analysis showed that errors were largest in the highest actual-strength quartile and that the 8–28 day age group had the largest age-group MAE. Cement content showed a moderate positive predictive association with strength, but this does not establish the causal result of changing cement in practice. Controlled mixture experiments would be required for causal conclusions.